

Title: Explainable AI Framework for Loan Default Prediction Using Ensemble Machine Learning (BATCH – 2)

Abstract

Financial institutions process a large number of loan applications every day, making accurate credit risk assessment essential for minimizing financial losses and ensuring responsible lending. Traditional loan approval methods often rely on manual evaluation and rule-based decision-making, which may fail to capture complex relationships among applicant attributes and provide limited transparency regarding prediction outcomes. This project proposes an Explainable Machine Learning-Based Loan Default Prediction System that leverages supervised machine learning algorithms to predict whether a loan applicant is likely to default while providing interpretable explanations for the model's decisions.

The proposed system utilizes historical loan application data containing applicant demographic information, financial details, employment status, credit history, loan characteristics, and repayment records. The dataset undergoes preprocessing techniques including missing value handling, categorical feature encoding, feature scaling, outlier treatment, and feature selection to improve data quality and model performance. Multiple machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, XGBoost, and Support Vector Machine are trained and evaluated to identify the most effective predictive model for loan default classification.

To enhance model transparency and user trust, Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are incorporated to identify the key features influencing each prediction. These explanations enable financial analysts and loan officers to understand the reasoning behind model decisions, ensuring fairness, accountability, and regulatory compliance in the loan approval process.

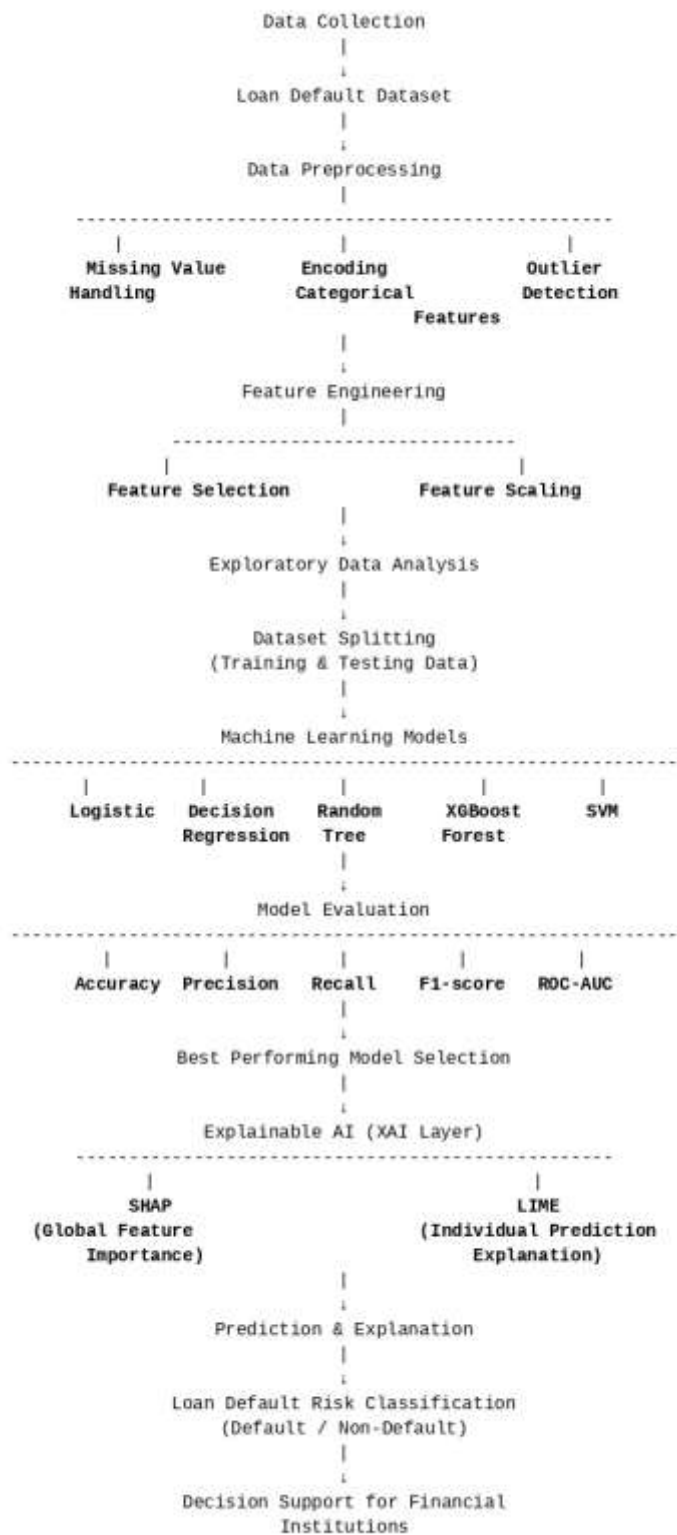
The performance of the proposed system is evaluated using accuracy, precision, recall, F1-score, ROC-AUC score, confusion matrix, and cross-validation techniques to ensure robust and reliable predictions. The resulting framework aims to provide accurate loan default prediction while delivering transparent, explainable, and data-driven insights that assist financial institutions in reducing credit risk, improving loan approval decisions, minimizing default rates, and enhancing overall financial decision-making.

Datasets

Dataset	URL link
Give Me Some Credit Dataset	https://raw.githubusercontent.com/aashish-bidap/Give-Me-Some-Credit/master/cs-training.csv

LendingClub Loan Default Dataset	https://www.kaggle.com/datasets/wordsforthewise/lending-club
Credit Risk Dataset – 50K Loans	https://zenodo.org/records/20830370?utm_source=chatgpt.com

Approach



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